minus first law. In other words, states change so that information and distinctions are never erased. In classical mechanics, this principle led to Hamilton's equations and Liouville's theorem. Earlier, in Lecture 4, I explained how in quantum mechanics this law led to the principle of unitarity, which in turn led to the general Schrödinger equation.

Lecture 8 was all about the first question: How do we describe the state of a particle? Now, in the current lecture, we come to the second question, which we might rephrase: *How do particles move in quantum mechanics?*

In Lecture 4, I laid out the basic rules for how quantum states change with time. The essential ingredient is the Hamiltonian H, which in both classical and quantum mechanics represents the total energy of a system. In quantum mechanics, the Hamiltonian controls the time evolution of a system through the time-dependent Schrödinger equation:

$$i\hbar\frac{\partial|\Psi\rangle}{\partial t} = \mathbf{H}|\Psi\rangle. \qquad (9.1)$$

This lecture is all about the *Original Schrödinger Equation*—the equation that Schrödinger wrote down to describe a quantum mechanical particle. The Original Schrödinger Equation is a special case of Eq. 9.1.

The motion of ordinary (nonrelativistic) particles in classical mechanics is governed by a Hamiltonian, equal to the kinetic energy plus the potential energy. We will soon come to the quantum version of this Hamiltonian, but first let's look at a Hamiltonian that's even simpler.

We'll start with the simplest Hamiltonian I can think of. In this case, the Hamiltonian operator **H** is a fixed constant times the momentum operator **P**:

$$\mathbf{H} = c\mathbf{P}. \qquad (9.2)$$

This example is rarely written down, though it turns out to be quite instructive. The constant c is a fixed number. Is $c\mathbf{P}$ a reasonable Hamiltonian for a particle? Yes it is, and in a moment we'll find out what kind of particle it describes. For now, just notice that Eq. 9.2 is different from what we might